MTH107 2021S2 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ENG301 Microprocessor Programming
* ENG355 Machine Learning
* ICT101 Computational Thinking
* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation
* MTH356 Advanced Mathematical Optimisation
* MTH366 Fundamentals of Graph Theory
* MTH367 Network Optimisation and Modelling

Donaldson Tan

MTH107

Timed Online Assignment – July Semester 2021

Calculus I

Tuesday, 16 November 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

<u>Without using L'Hopital's Rule</u>, determine the following limits.

(a) Determine the following limit. Show your steps clearly.

$$\lim_{x \to 1} \frac{(x^2 - 1)^2}{x - 1}$$

(6 marks)

(b) Determine the following limit. Show your steps clearly.

$$\lim_{x \to \infty} (\sqrt{x^2 + x} - x)$$

(8 marks)

(c) Let f be a function defined on $(0,3)\backslash\{1\}$. That is, f is defined on $(0,3)$, except possibly at 1.

Determine $\lim_{x \to 1} f(x)$, given that $4x - 2 \leq f(x) \leq 2x^2$ for all x in the interval $(0,3)$.

(6 marks)

Question 2

(a) Let

$$f(x) = \begin{cases} a - x & if\ x \leq \frac{\pi}{2} \\ a\cos x & if\ x > \frac{\pi}{2} \end{cases}.$$

Is it possible to choose a suitable value of a such that f is continuous? Justify your answer.

(7 marks)

(b) Let

$$f(x) = \begin{cases} x + 5 & if\ x < 1 \\ 4 + 2\sqrt{x} & if\ x \geq 1 \end{cases}.$$

Is f continuous at 1? Justify your answer.

(5 marks)

(c) Let f be the same function as defined in Question 2(b).
Is f differentiable at $x = 1$? If yes, determine the value of $f'(1)$. Justify your answer.

(8 marks)

Question 3

(a) Let

$$y = \frac{\sqrt{x^2+2} - \sqrt{x^2-2}}{\sqrt{x^2+2} + \sqrt{x^2-2}}.$$

Find $\frac{dy}{dx}$ and express your answer in the form $x - \frac{ax^3}{\sqrt{x^4-b}}$, where a and b are constants to be determined.

(10 marks)

(b) Let $f: [a, b] \to \mathbb{R}$ be a differentiable function such that $f(a) = f(b) = 0$. Show that the equation $f'(x) = 0$ has a solution c satisfying $a < c < b$.

(10 marks)

Question 4

(a) Find the Taylor's series for $\ln(1 + x^2)$, about $x = 0$. Write your answer up to the first 4 nonzero terms.

Determine the 2021th term of the Taylor series.

(8 marks)

(b) Find the Taylor's series of $\frac{e^x - 1}{x}$ (about $x = 0$), writing your answer up to the first 4 nonzero terms.

Use the above Taylor series to find the exact value of the following series:

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \frac{5}{6!} + \cdots$$

(12 marks)

Question 5

Let f be the function defined on $(-\infty, \infty)$ by

$$f(x) = \frac{3x^2}{x^2 + 9}.$$

(a) Compute the interval on which f is increasing.

(5 marks)

(b) Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(5 marks)

(c) Apply derivative tests to determine the critical point(s) of f and discuss their nature. Show your steps clearly.

(5 marks)

(d) Sketch the graph of f.

(5 marks)

----- END OF PAPER -----

1(a) $\lim_{x \to 1} \frac{(x^2-1)^2}{x-1} = \lim_{x \to 1} \frac{(x-1)^2(x+1)^2}{x-1}$

$= \lim_{x \to 1} (x-1)(x+1)^2$

$= 0$

(b) $\lim_{x \to \infty} \left(\sqrt{x^2+x} - x\right) = \lim_{x \to \infty} \frac{\sqrt{x^2+x} - x}{1} \times \frac{\sqrt{x^2+x} + x}{\sqrt{x^2+x} + x}$

$= \lim_{x \to \infty} \frac{(x^2+x) - x^2}{\sqrt{x^2+x} + x}$

$= \lim_{x \to \infty} \frac{x}{\sqrt{x^2+x} + x}$

$= \lim_{x \to \infty} \frac{1}{\sqrt{\frac{x^2+x}{x^2}} + 1}$

$= \lim_{x \to \infty} \frac{1}{\sqrt{1+\frac{1}{x}} + 1}$

$= \frac{1}{1+1}$

$= \frac{1}{2}$

(c) $4x-2 \leq f(x) \leq 2x^2$

$\lim_{x \to 1} (4x-2) \leq \lim_{x \to 1} f(x) \leq \lim_{x \to 1} 2x^2$

$2 \leq \lim_{x \to 1} f(x) \leq 2$

By Squeeze Theorem

$\therefore \lim_{x \to 1} f(x) = 2$

2(a) suppose $f(x)$ is continuous at $x=\frac{\pi}{2}$

$$f\left(\frac{\pi}{2}\right) = \lim_{x\to\frac{\pi}{2}^+} f(x)$$

$$a-\frac{\pi}{2} = a\cos\frac{\pi}{2} = 0$$

$$\therefore a = \frac{\pi}{2}$$

$\therefore$ $f(x)$ is continuous when $a=\frac{\pi}{2}$

(b) $f(1) = 4+2\sqrt{1} = 6$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x+5) = 6$$

since $f(1) = \lim_{x\to 1^-} f(x)$, then $f(x)$ is continuous at $x=1$

(c)
$$\lim_{h\to 0^+} \frac{f(h+1)-f(1)}{h} = \lim_{h\to 0^+} \frac{(4+2\sqrt{h+1})-6}{h}$$

$$= \lim_{h\to 0^+} \frac{2\sqrt{h+1}-2}{h}$$

$$= \lim_{h\to 0^+} \frac{2}{h}\left(\sqrt{h+1}-1\right)$$

use Maclaurin series for $(1+x)^k$. where $h=x$ and $k=\frac{1}{2}$

$$= \lim_{h\to 0^+} \frac{2}{h}\left[\sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}h^n - 1\right]$$

$$= \lim_{h\to 0^+} \frac{2}{h}\left[\frac{1}{2}h - \frac{1}{8}h^2 + \frac{1}{8}h^3 + \dots\right]$$

$$= \lim_{h\to 0^+} \left(1 - \frac{1}{4}h + \frac{1}{4}h^2 + \dots\right)$$

$$= 1$$

$$\lim_{h\to 0^-} \frac{f(h+1)-f(1)}{h} = \lim_{h\to 0^-} \frac{(h+1)+5)-6}{h}$$

$$= \lim_{h\to 0^-} \frac{h}{h}$$

$$= 1$$

$$\lim_{h\to 0^+} \frac{f(h+1)-f(1)}{h} = \lim_{h\to 0^-} \frac{f(h+1)-f(1)}{h} = 1 \Rightarrow \lim_{h\to 0} \frac{f(h+1)-f(1)}{h} = 1$$

$\therefore$ $f'(1)$ exists. $f'(1) = \lim_{h\to 0} \frac{f(h+1)-f(1)}{h} = 1$

3(a)

$$y = \frac{\sqrt{x^2+2} - \sqrt{x^2-2}}{\sqrt{x^2+2} + \sqrt{x^2-2}} \times \frac{\sqrt{x^2+2} - \sqrt{x^2-2}}{\sqrt{x^2+2} - \sqrt{x^2-2}}$$

$$= \frac{(\sqrt{x^2+2} - \sqrt{x^2-2})^2}{(x^2+2) - (x^2-2)}$$

$$= \frac{(x^2+2) - 2\sqrt{x^4-4} + (x^2-2)}{4}$$

$$= \frac{2x^2 - 2\sqrt{x^4-4}}{4}$$

$$= \frac{1}{2}x^2 - \frac{1}{2}\sqrt{x^4-4}$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{2}x^2 - \frac{1}{2}\sqrt{x^4-4}\right)$$

$$= x - \frac{1}{2}\left(\frac{1}{2}\right)(x^4-4)^{-\frac{1}{2}}(4x^3)$$

$$= x - \frac{x^3}{\sqrt{x^4-4}}$$

$$\therefore a = 1,\ b = 4$$

(b) By Mean Value Theorem, $\exists c$ on (a,b) such that

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

$$= \frac{0-0}{b-a}$$

$$= 0$$

4(a) $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$

$$\ln(1+x^2) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{n}$$

$$= x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} + \dots$$

$$2021^{th} \text{ term} = \frac{x^{2(2021)}}{2021} = \frac{x^{4042}}{2021}$$

(b) $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$

$$\frac{e^x - 1}{x} = \frac{1}{x}(e^x - 1)$$

$$= \frac{1}{x}\left(\sum_{n=0}^{\infty} \frac{x^n}{n!} - 1\right)$$

$$= \frac{1}{x}\left(\frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\right)$$

$$= \frac{1}{1!} + \frac{x}{2!} + \frac{x^2}{3!} + \frac{x^3}{4!} + \dots$$

let $f(x) = \frac{e^x - 1}{x}$

$$f'(x) = e^x x^{-1} - x^{-2}(e^x - 1)$$

$$f'(x) = \frac{d}{dx}\left[\frac{1}{1!} + \frac{x}{2!} + \frac{x^2}{3!} + \frac{x^3}{4!} + \frac{x^4}{5!} + \frac{x^5}{6!} + \dots\right]$$

$$= \frac{1}{2!} + \frac{2x}{3!} + \frac{3x^2}{4!} + \frac{4x^3}{5!} + \frac{5x^4}{6!} + \dots$$

$$\text{at } x = 1: \quad \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \frac{5}{6!} + \dots$$

$$= f'(1)$$

$$= e^1 (1)^{-1} - 1^{-2}(e - 1)$$

$$= e - (e - 1)$$

$$= 1$$

5(a) $f(x) = \frac{3x^2}{x^2+9} = \frac{3(x^2+9)-27}{x^2+9} = 3 - \frac{27}{x^2+9}$

$f'(x) = -27(-1)(x^2+9)^{-2}(2x)$

$= \frac{54x}{(x^2+9)^2}$

let $f'(x) > 0$

$\Rightarrow 54x > 0$

$\therefore x > 0$

$f(x)$ is increasing when $x > 0$

(b) $\min(x^2+9) = 9$

$\Rightarrow$ there is no vertical asymptote.

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left[3 - \frac{27}{x^2+9}\right] = 3$

$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty}\left[3 - \frac{27}{x^2+9}\right] = 3$

$y = 3$ is the horizontal asymptote

(c) let $f'(x) = 0$

$54x = 0$

$x = 0$

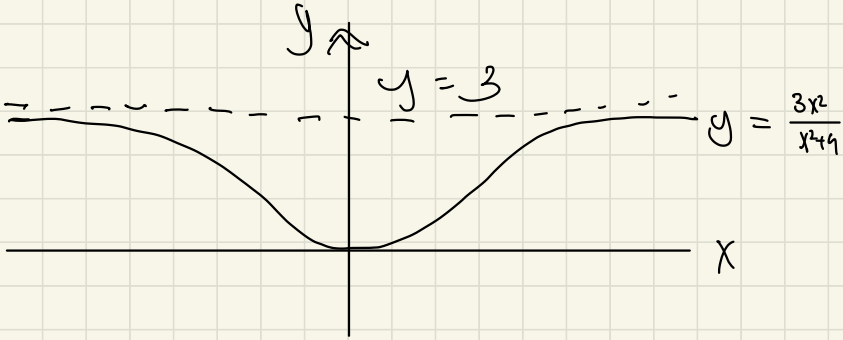
$f(0) = 3 - \frac{27}{9} = 3$

$\therefore (0,0)$ is a critical point

$f''(x) = 54(x^2+9)^{-2} + 54x(-2)(x^2+9)^{-3}(2x)$

$= \frac{54}{(x^2+9)^2} - \frac{216x}{(x^2+9)^3}$

$f''(0) = \frac{54}{9^2} - 0 = \frac{2}{3}$

$\therefore (0,0)$ is a minima

5(c)